





Management and Curation of EarthPortal Resources

Two observations on the Metadata Guideline and the EarthPortal schema

Student

Emmanuel Nyandu Kagarabi

Internship Supervisor

Christelle Pierkot

14 October 2026

1 Introduction

In this short report, we highlight a few observations we made while developing our pipeline on the state of the art of the semantic artefacts hosted on EarthPortal. Even though we will discuss that pipeline at the end of October, please *think about our two concerns* related to the Metadata Guideline:

1. *some properties of the guideline (MOD 3) do not have the same name in the metadata schema of EarthPortal (MOD 1.4), and five of them are not linked to it at all*;
2. *some fields of the metadata schema are expected by the portal, and most of them by O'FAIRe, but they are not in the guideline.*

The guideline is taken in its machine-readable form, the cheatsheet of Report 1 (99 properties). The metadata schema is the one the portal publishes through its API[1].

2 Development

2.1 Observation 1: some properties are stored under another name

The cheatsheet stores 99 properties: 20 are Mandatory, 58 Recommended and 21 Optional. In notebook 04[2], each of them is linked to a field of the metadata schema with one of three rules (Table 1).

Rule	Meaning	Properties
same	The portal has a field with exactly the same name.	72
mapping	The portal has a field with another name, and this field lists the property of the cheatsheet among its synonyms (`metadataMappings`).	21
manual	No rule above works; we wrote the link by hand in `reference/manual_links.csv`.	6
	Total	**99**

Table 1: How the 99 properties of the cheatsheet are linked to the EarthPortal schema.

Figure 1 shows how a synonym is declared. The field `omv:description` (Figure 1a) lists six properties in `metadataMappings`; so, when an ontology file describes itself with `dcterms:description` or `schema:description`, the portal stores the value in `omv:description`. For Description, the cheatsheet already uses `omv:description`, so it is a *same* case; but Keywords works exactly in this way: the cheatsheet says `dcat:keyword`, and the portal field `omv:keywords` lists `dcat:keyword` among its synonyms (Figure 1b), so it is a *mapping* case.

[1] https://data.earthportal.eu/submission_metadata (110 fields) and https://data.earthportal.eu/ontology_metadata (20 fields).

[2] https://github.com/EarthPortal/SA_Management/tree/main/earthportal_sa_audit

```
{
  "@id": "http://data.bioontology.org/submission_metadata/description",
  "@type": "http://data.bioontology.org/metadata/SubmissionMetadata",
  "attribute": "description",
  "namespace": "omv",
  "label": "Description",
  "extracted": true,
  "metadataMappings": [
    "omv:description",
    "dc:description",
    "dcterms:description",
    "doap:description",
    "schema:description",
    "oboInOwl:remark"
  ],
```


(a) `omv:description` (same case)


```
{
  "@id": "http://data.bioontology.org/submission_metadata/keywords",
  "@type": "http://data.bioontology.org/metadata/SubmissionMetadata",
  "attribute": "keywords",
  "namespace": "omv",
  "label": "Keywords",
  "extracted": "once",
  "metadataMappings": [
    "omv:keywords",
    "mod:keyword",
    "dcat:keyword",
    "schema:keywords"
  ],
```


(b) `omv:keywords` (mapping case)

Figure 1: Equivalence between the properties of the cheatsheet and the fields of the EarthPortal schema. The list `metadataMappings` gives the properties the portal accepts as synonyms.

Figure 2 lists the 26 properties of the cheatsheet that the portal stores under another name. Twenty-one of them are *mapping* cases: the portal already knows the name used by the cheatsheet, so nothing has to change for them. The other five are *manual* cases (the sixth manual case, Categories, has the same name as the portal field; it is manual only because `omv:hasDomain` exists on both the ontology record and the submission).

```
{'same': 72, 'mapping': 21, 'manual': 6}
26 attributes are stored by the portal under another property
```

	id	entry	usage	property	portal_attribute	link_rule
0	1.1	Acronym	Mandatory	mod:acronym	omv:acronym	manual
8	1.9	Version IRI	Recommended	omv:versionIRI	owl:versionIRI	manual
19	2.6	Keywords	Recommended	dcat:keyword	omv:keywords	mapping
20	2.7	Subjects	Recommended	dct:subject	omv:hasDomain	mapping
25	3.2	Modification date	Mandatory	dct:modified	omv:modificationDate	mapping
28	3.5	Submission date	Mandatory	dct:dateSubmitted	omv:creationDate	mapping
35	5.2	Creator	Mandatory	dct:creator	omv:hasCreator	mapping
36	5.3	Contributor	Recommended	dct:contributor	omv:hasContributor	mapping
40	5.7	Endorsed by	Recommended	mod:endorsedBy	omv:endorsedBy	mapping
55	8.1	Bug database	Recommended	doap:bug-database	doap:bugDatabase	mapping
58	8.4	Mailing list	Recommended	doap:mailing-list	doap:mailingList	mapping
59	8.5	To do list	Recommended	mod:toDoList	voaf:toDoList	mapping
62	9.2	Design for task	Recommended	mod:designedForTask	omv:designedForOntologyTask	mapping
65	10.1	This ontology is a view of	Recommended	dct:isPartOf	bpm:viewOf	manual
68	10.4	Similar to	Recommended	mod:similar	door:similarTo	mapping
69	10.5	Generally related to	Optional	mod:semanticArtefactRelation	door:ontologyRelatedTo	manual
70	10.6	Comes from the same domain	Optional	mod:comesFromTheSameDomain	door:comesFromTheSameDomain	mapping
71	10.7	Has equivalences with	Recommended	mod:hasEquivalencesWith	door:isAlignedTo	mapping
72	10.8	Backward compatible	Optional	owl:backwardCompatibleWith	omv:isBackwardCompatibleWith	mapping
73	10.9	Incompatible	Optional	owl:incompatibleWith	omv:isIncompatibleWith	mapping
74	10.10	Disparate modelling with	Recommended	mod:hasDisparateModelling	door:hasDisparateModelling	manual
75	10.11	Has disjunctions with	Optional	mod:hasDisjunctionsWith	voaf:hasDisjunctionsWith	mapping
76	10.12	Generalization of	Optional	mod:generalizes	voaf:generalizes	mapping
77	10.13	Specialization of	Optional	mod:specializes	door:explanationEvolution	mapping
78	10.14	Imports	Recommended	owl:imports	omv:useImports	mapping
79	10.15	Used by	Recommended	mod:usedBy	voaf:usedBy	mapping

Figure 2: Output of notebook 04: the 26 properties of the cheatsheet that the portal stores under another name, with the rule that links them. For more details on the manual cases, click here.

Table 2 gives these five properties. For each of them, the portal field has no synonym that matches the cheatsheet. So, if an author follows the guideline and writes, for example, `mod:semanticArtefactRelation` in the ontology file, we expect the portal not to recognise it. To understand where each name comes from, we also looked at the current version of MOD (3.2.5)[3].

Id	Entry	Cheatsheet	EarthPortal	What we found
1.1	Acronym	`mod:acronym`	`omv:acronym` (ontology record)	MOD 3.2 declares `mod:acronym`; the portal field has no synonym.
1.9	Version IRI	`omv:versionIRI`	`owl:versionIRI`	MOD 3.2 uses `owl:versionIRI`; we did not find `omv:versionIRI` in MOD. Here, the guideline may be the one to correct.
10.1	This ontology is a view of	`dct:isPartOf`	`bpm:viewOf` (ontology record)	The portal has no field for `dct:isPartOf`; it has the inverse, `dct:hasPart` ("Has part (has views)").
10.5	Generally related to	`mod:semanticArtefactRelation`	`door:ontologyRelatedTo`	MOD 3.2 declares both properties equivalent, but the portal field does not list the MOD name.
10.10	Disparate modelling with	`mod:hasDisparateModelling`	`door:hasDisparateModelling`	The portal lists `mod:hasDisparateModellingWith`, while `mod.ttl` declares `mod:hasDisparateModelling`: one spelling has to be aligned.

Table 2: The five properties of the cheatsheet that are not linked to the EarthPortal schema, neither by name nor by a synonym.

Before the publication of this guideline, it could be better for the development team to update the code and declare new mappings for these five properties, or to agree with us on the name the guideline should use. The 21 mapped properties need nothing.

2.2 Observation 2: some fields expected by the portal are not in the guideline

We then looked at the other direction: the fields of the metadata schema that no property of the guideline points to. Of the 130 fields (110 on the submission, 20 on the ontology record), 31 are not used by the guideline:

- 21 are managed by the portal itself: they have no label, their category is "no", and they are not extracted from the ontology file (for example `submissionId`, `submissionStatus`, `uploadFilePath`, `metrics`, `notes`, `reviews`, `views`). They do not belong in a guideline for authors;
- 10 have a label in the submission form and are extracted from the ontology file. They are the candidates of Table 3.

The last two columns of Table 3 say whether O'FAIRe reads the field (with the question and its points), and in how many of the 68 artefacts (views and metadata-only records excluded) the latest submission has a value today.

[3] File `mod.ttl` of https://github.com/FAIR-IMPACT/MOD, branch `main`, read on 14 October 2026.

Portal field	Label in the portal	Category	O'FAIRe question (points)	Filled
`omv:version`	Version information	general	R1.2Q4 (4)	56 / 68
`owl:deprecated`	Deprecated	general	A2Q4 (4)	55 / 68
`bpm:prefLabelProperty`	Object preferred label property	object	I3Q3 (7), R1Q1 (5)	55 / 68
`bpm:synonymProperty`	Object synonym property	object	I3Q3 (7), R1Q1 (5)	55 / 68
`bpm:definitionProperty`	Object definition property	object	I3Q3 (7), R1Q1 (5)	55 / 68
`bpm:authorProperty`	Object author property	object	I3Q3 (7), R1Q1 (5)	55 / 68
`bpm:obsoleteProperty`	Object obsolete property	object	I3Q3 (7), R1Q1 (5)	55 / 68
`bpm:hierarchyProperty`	Hierarchy property	object	I3Q3 (7), R1Q2 (3)	57 / 68
`bpm:createdProperty`	Object creation date property	object	no	55 / 68
`bpm:modifiedProperty`	Object modification date property	object	no	55 / 68

Table 3: Fields of the EarthPortal schema that could be added to the guideline. "object" stands for the category "object description properties". In O'FAIRe, MOD 1.4 calls the first field `owl:versionInfo` and the object description fields `mod:...Property`.

Three remarks on this table:

1. Eight of the ten fields bring points in O'FAIRe, yet an author who follows the guideline has no way to know that they exist. The guideline already has Version IRI (1.9) and Prior version (10.2), but not the version number itself; and it already has Root of obsolete branch (11.1, `bpm:obsoleteParent`), which describes the content in the same way as the object description fields.
2. The values look like default values rather than choices of the authors. In the 55 filled submissions, each object description field has exactly one value: for instance, `prefLabelProperty` is `skos:prefLabel` everywhere, including in 23 OWL artefacts, and `deprecated` is `false` everywhere. We have not checked where these defaults come from.
3. Some of these values are not in the list of values the portal itself proposes (`enforcedValues`). For 22 OWL artefacts, `hierarchyProperty` is `owl:subClassOf`, while the list proposes `rdfs:subClassOf`; and `createdProperty` and `modifiedProperty` are `dc:created` and `dc:modified` (Dublin Core Elements 1.1), while the list proposes `dcterms:created` and `dcterms:modified`.

The ten fields of Table 3 could be added to the guideline, at least the eight that O'FAIRe reads. If they are added, the guideline could also give the allowed values, so that authors replace the defaults with the properties their artefact really uses.

3 Conclusion

While I am still completing and building some intermediary steps within my pipeline (five Jupyter notebooks), I would appreciate it if you could check my two statements and provide some feedback:

1. five properties of the guideline are not linked to the EarthPortal schema (Table 2), and the development team could declare new mappings for them before the guideline is published;

2. ten fields of the schema, eight of them read by O'FAIRe, could be added to the guideline (Table 3).

Everything can be reproduced with notebook 04 of our repository[4].

[4] https://github.com/EarthPortal/SA_Management, folder `earthportal_sa_audit`.